

SMAP CDF-matching sensitivity experiment over North Morocco

Domain: NorthMor | Year: 2016

Objective:

This report evaluates the internal behaviour of the SMAP EnKF assimilation system over NorthMor in 2016. It focuses on SMAP observation availability, innovations, analysis increments, seasonal correction patterns, and ensemble spread diagnostics for the noCDF and CDF assimilation configurations.

Experiments Compared:

- OPL_noirr_2016
- DA_smmap_nocdf_noirr_2016
- DA_smmap_cdf_noirr_2016

Note: This report does not include hydrological impact diagnostics. Soil moisture, ET, runoff, groundwater, streamflow, and external validation diagnostics were not generated in this report version.

Table of Contents

1. Report scope
2. DA-SMAP-noCDF assimilation diagnostics
3. DA-SMAP-CDF assimilation diagnostics
4. noCDF vs CDF interpretation notes
5. Surface soil moisture response
6. Root-zone and vertical soil moisture propagation
7. Surface fluxes
8. Groundwater response
9. Runoff and baseflow response
10. Streamflow & External validation
11. Scientific checklist

4. noCDF vs CDF interpretation notes

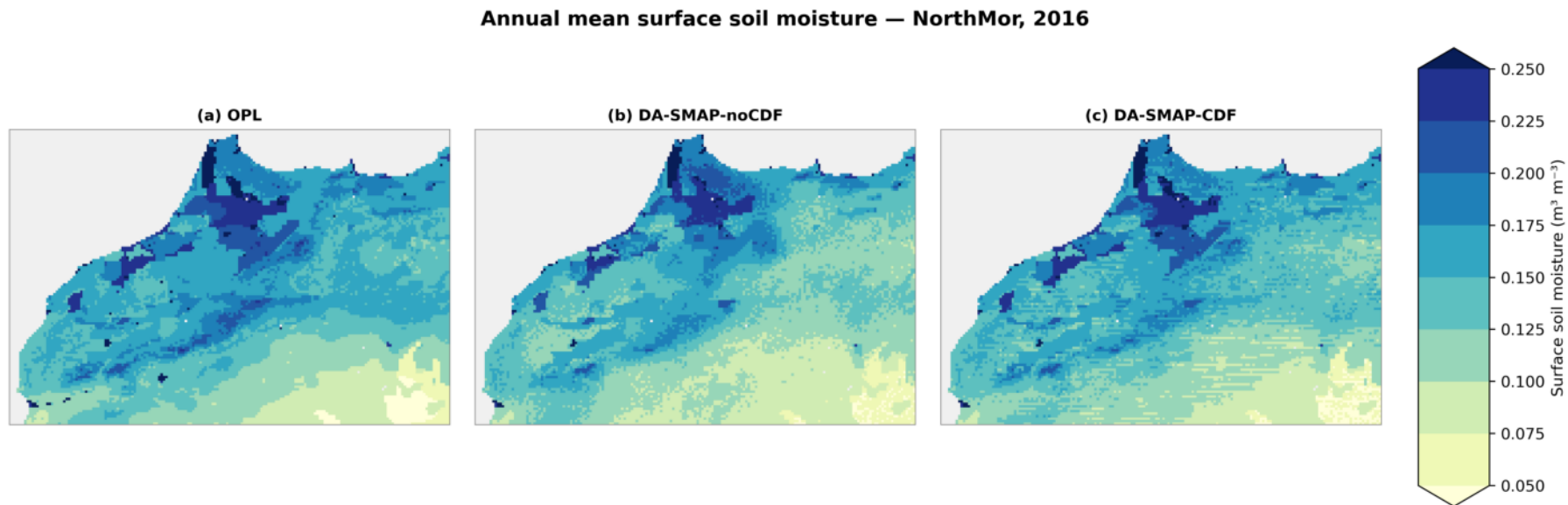
When comparing noCDF and CDF assimilation diagnostics, check:

- whether the spatial coverage of assimilated observations is consistent between experiments;
- whether CDF-matching reduces systematic innovation bias;
- whether the increment distribution becomes more centred around zero;
- whether wet-season and dry-season increments show coherent spatial patterns;
- whether spread diagnostics remain consistent;
- whether any difference in observation count reflects a real processing choice or a labelling/input-file issue.

Do not invent scientific conclusions. This page should only guide the interpretation.

5. Surface soil moisture response

Annual mean surface soil moisture — NorthMor, 2016

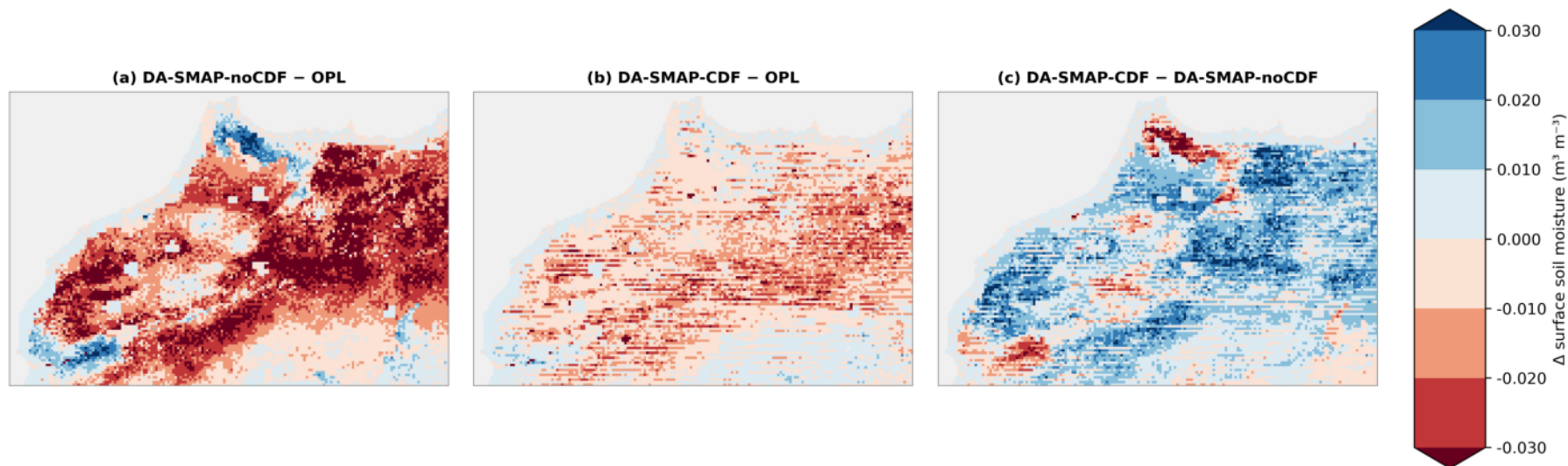


Source file: surface_soil_moisture_annual_mean_2016.png

Annual mean spatial distribution of surface soil moisture.

Surface soil moisture response to assimilation — 2016

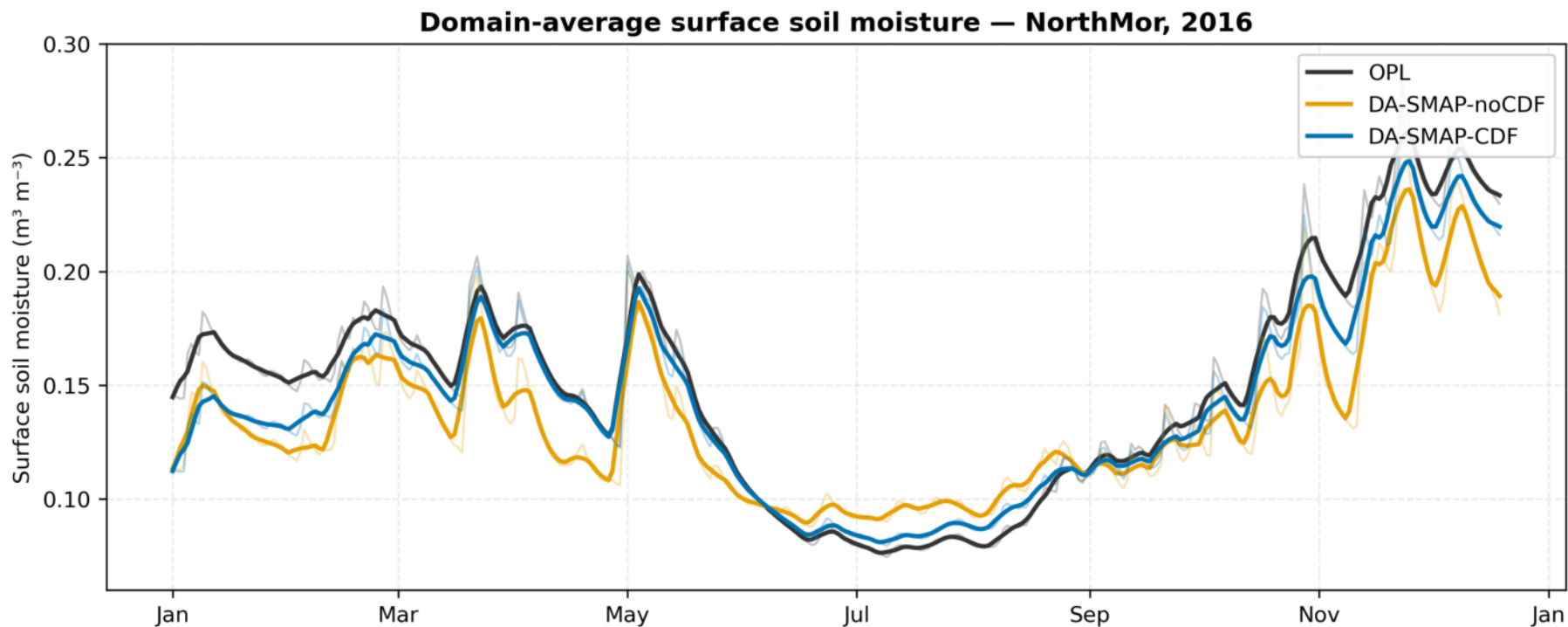
Surface soil moisture response to assimilation — 2016



Source file: surface_soil_moisture_differences_2016.png

Impact of SMAP assimilation on surface soil moisture.

Domain-average surface soil moisture time series — 2016

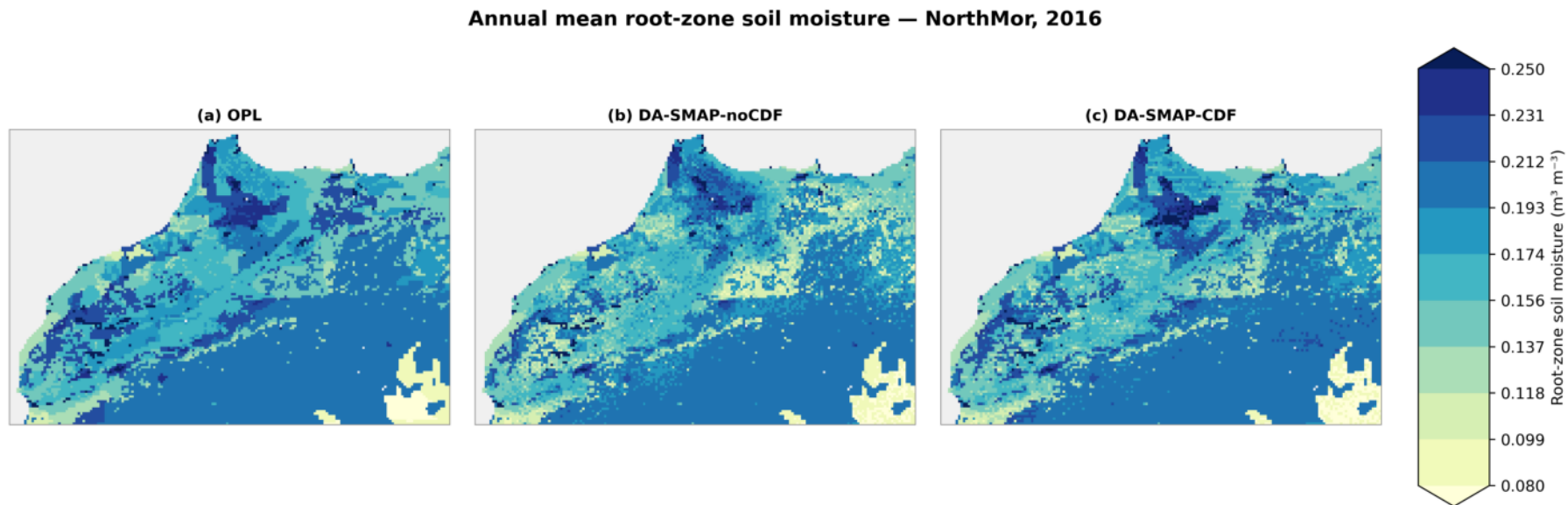


Source file: surface_soil_moisture_domain_mean_timeseries_2016.png

Daily domain-averaged timeseries of surface soil moisture.

6. Root-zone and vertical soil moisture propagation

Annual mean root-zone soil moisture — NorthMor, 2016

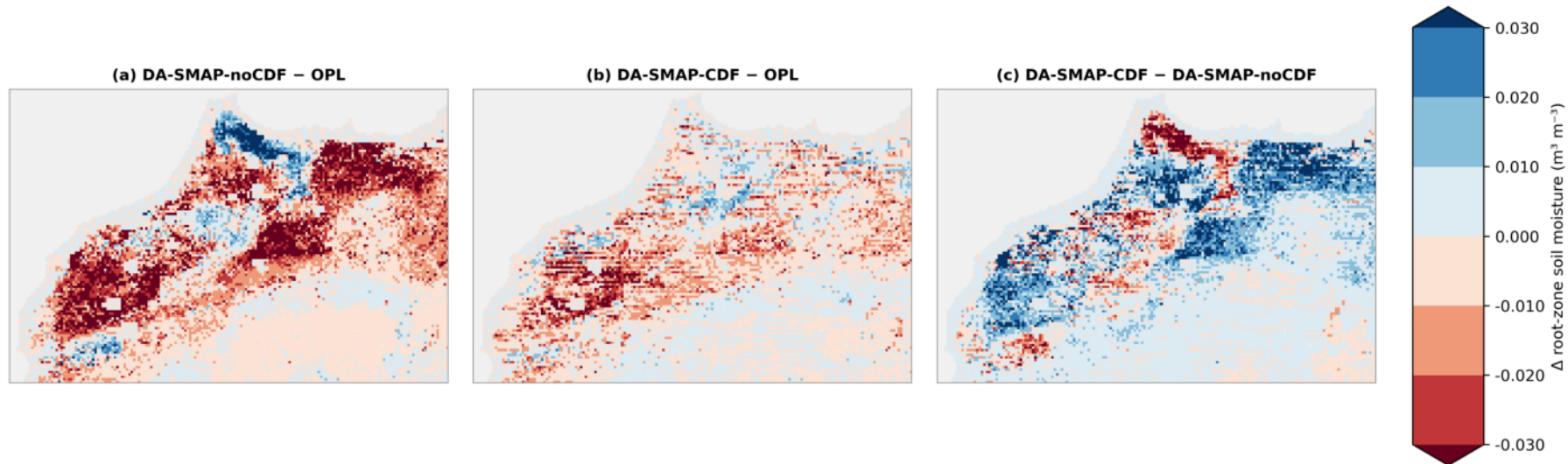


Source file: rootzone_soil_moisture_annual_mean_2016.png

Annual mean spatial distribution of root-zone soil moisture.

Root-zone soil moisture response to assimilation — 2016

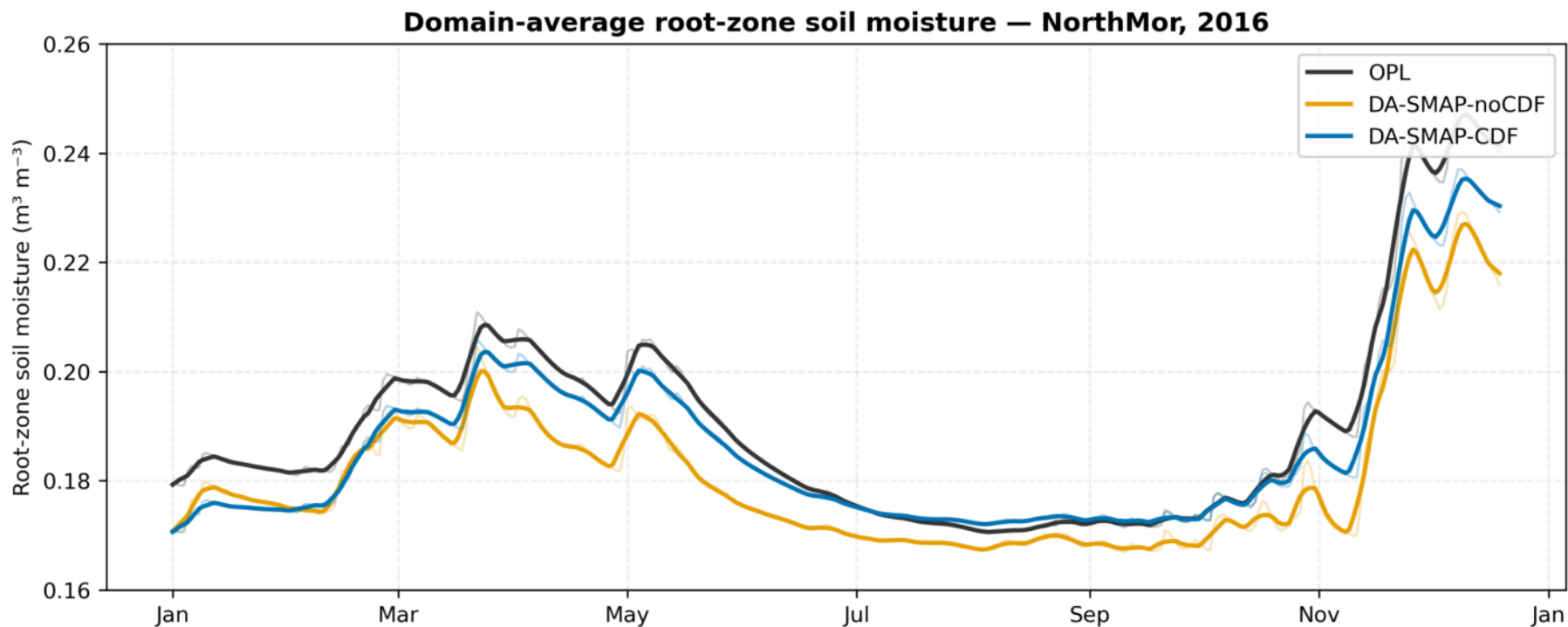
Root-zone soil moisture response to assimilation — 2016



Source file: rootzone_soil_moisture_differences_2016.png

Impact of SMAP assimilation on root-zone soil moisture.

Domain-average root-zone soil moisture time series — 2016

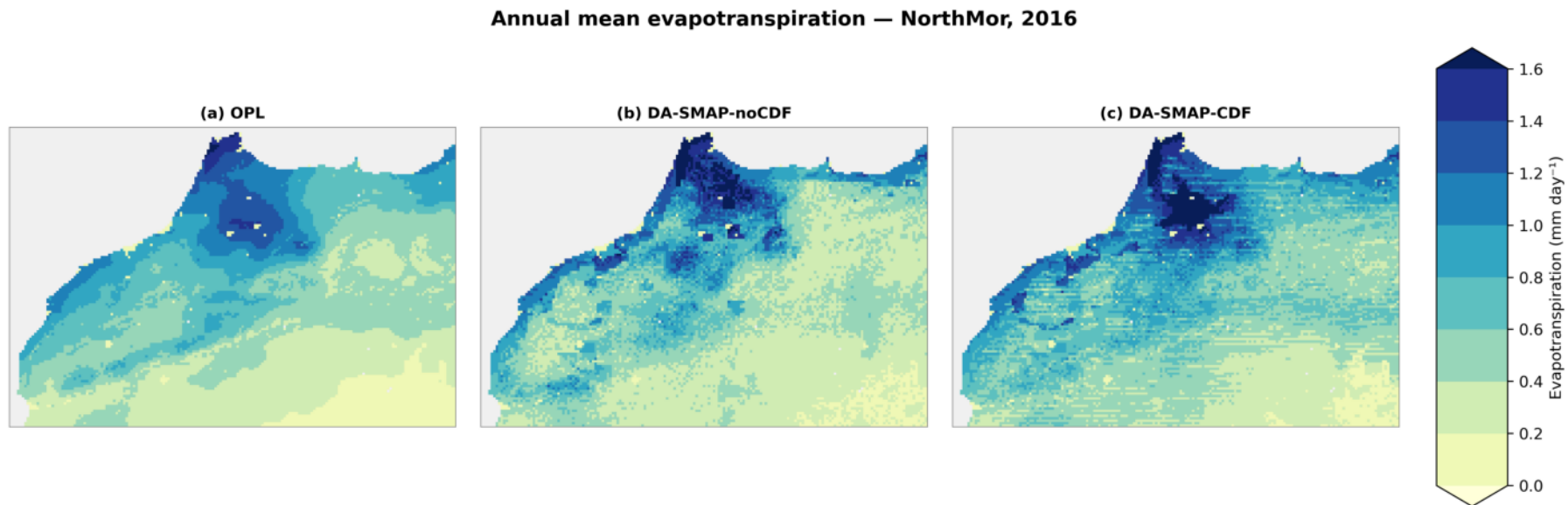


Source file: rootzone_soil_moisture_domain_mean_timeseries_2016.png

Daily domain-averaged timeseries of root-zone soil moisture.

7. Surface fluxes

Annual mean evapotranspiration — NorthMor, 2016

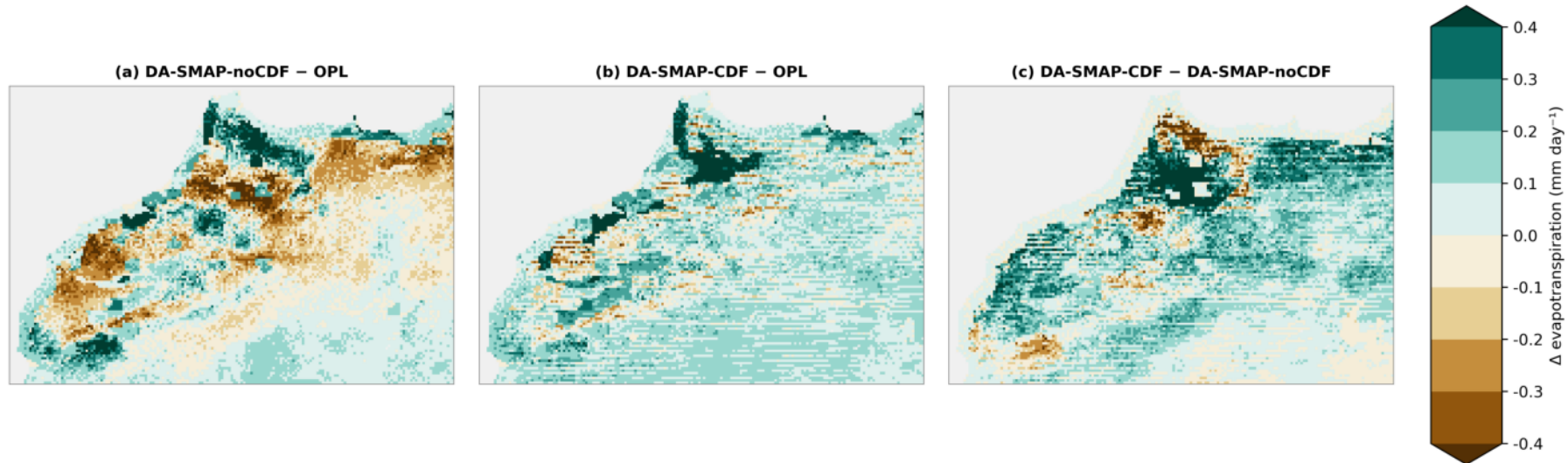


Source file: evapotranspiration_annual_mean_2016.png

Annual mean spatial distribution of evapotranspiration.

Evapotranspiration response to assimilation — 2016

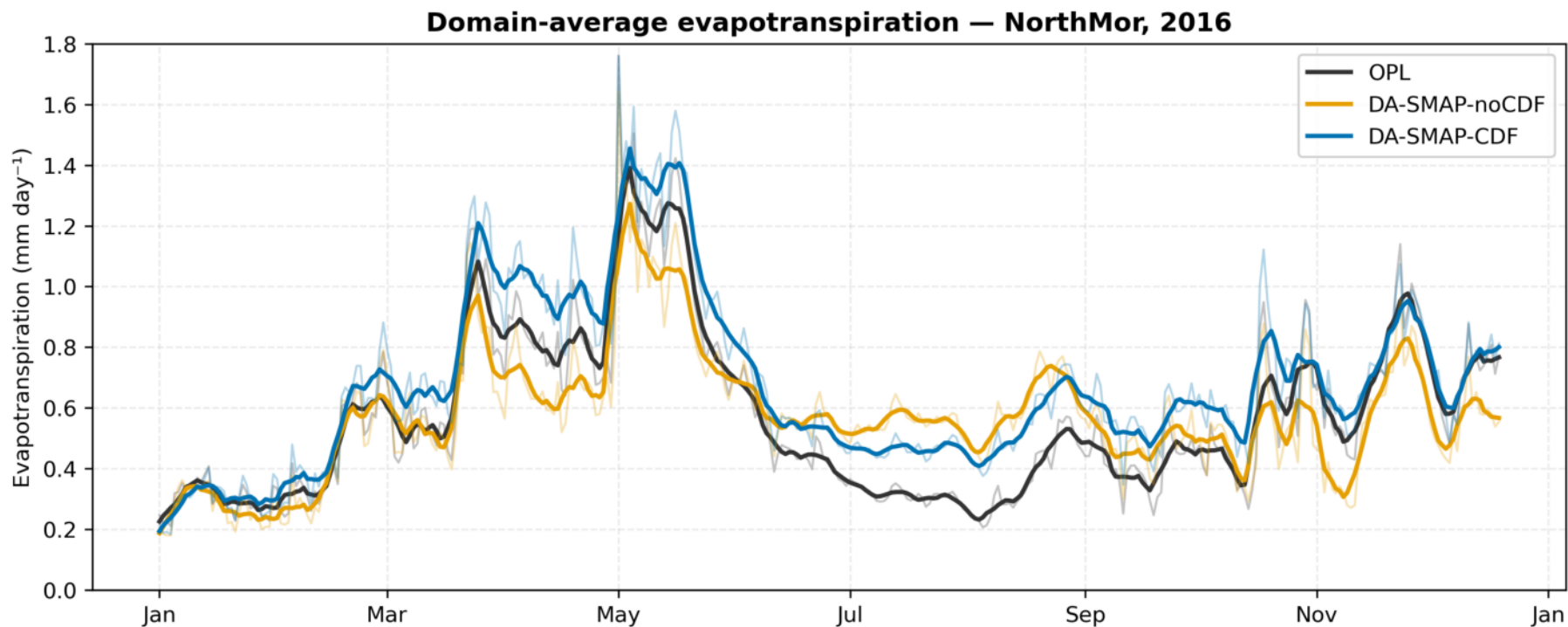
Evapotranspiration response to assimilation — 2016



Source file: evapotranspiration_differences_2016.png

Impact of SMAP assimilation on evapotranspiration.

Domain-average evapotranspiration time series — 2016

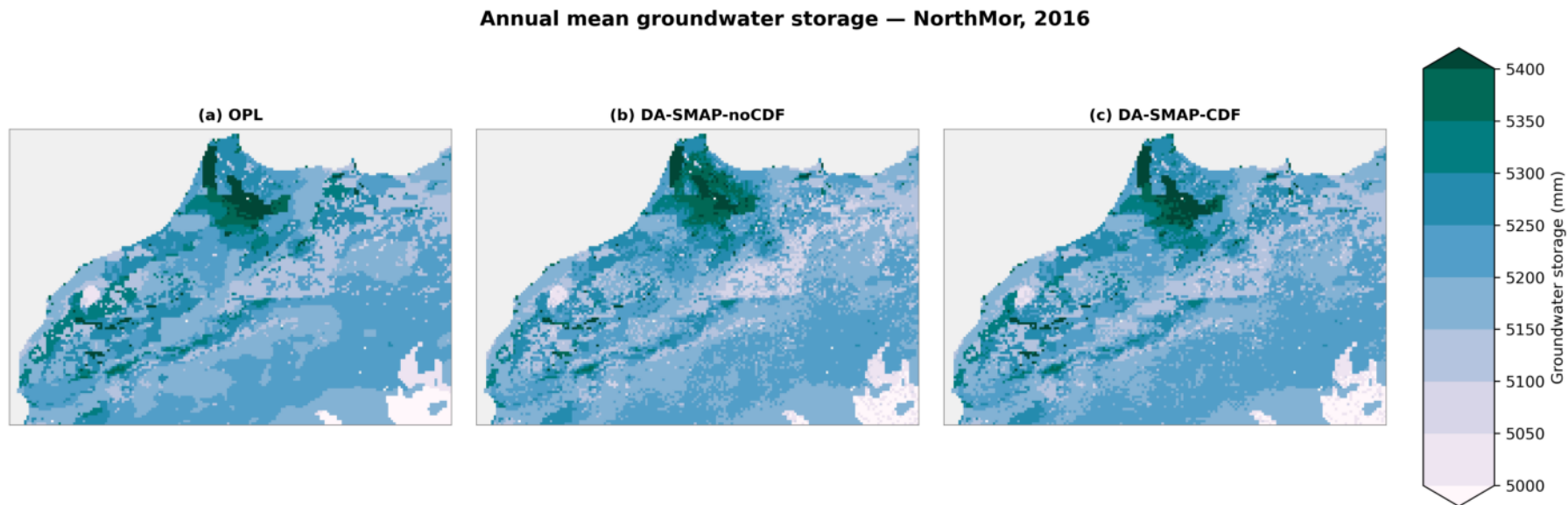


Source file: evapotranspiration_domain_mean_timeseries_2016.png

Daily domain-averaged timeseries of evapotranspiration.

8. Groundwater response

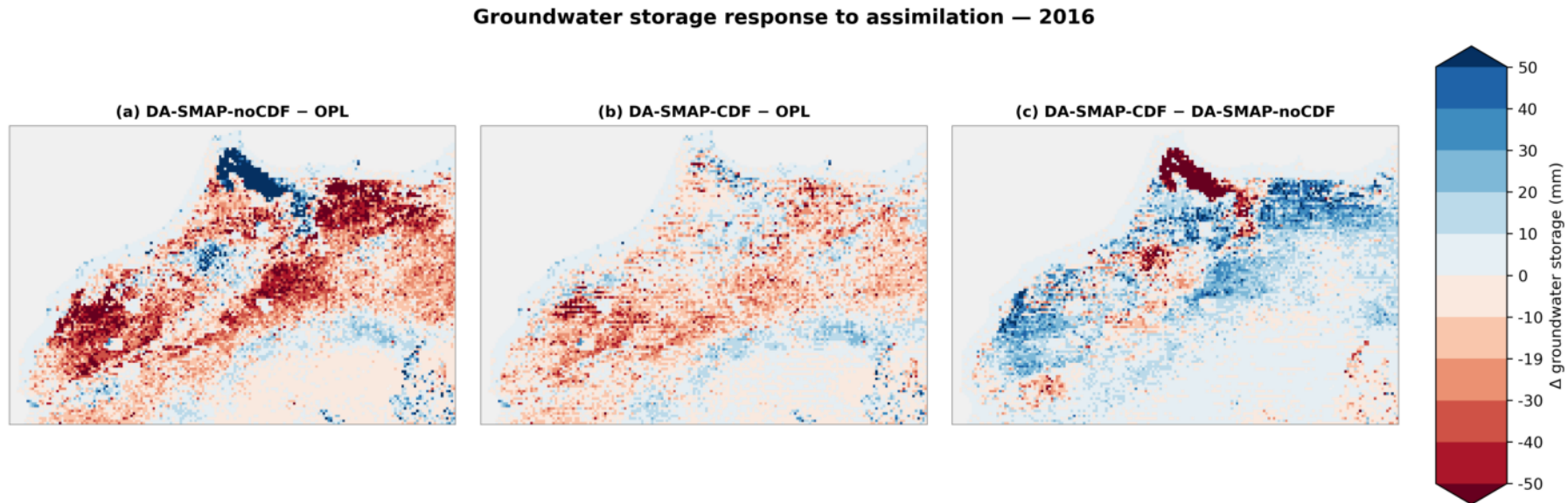
Annual mean groundwater storage — NorthMor, 2016



Source file: groundwater_storage_annual_mean_2016.png

Annual mean spatial distribution of groundwater storage.

Groundwater storage response to assimilation — 2016

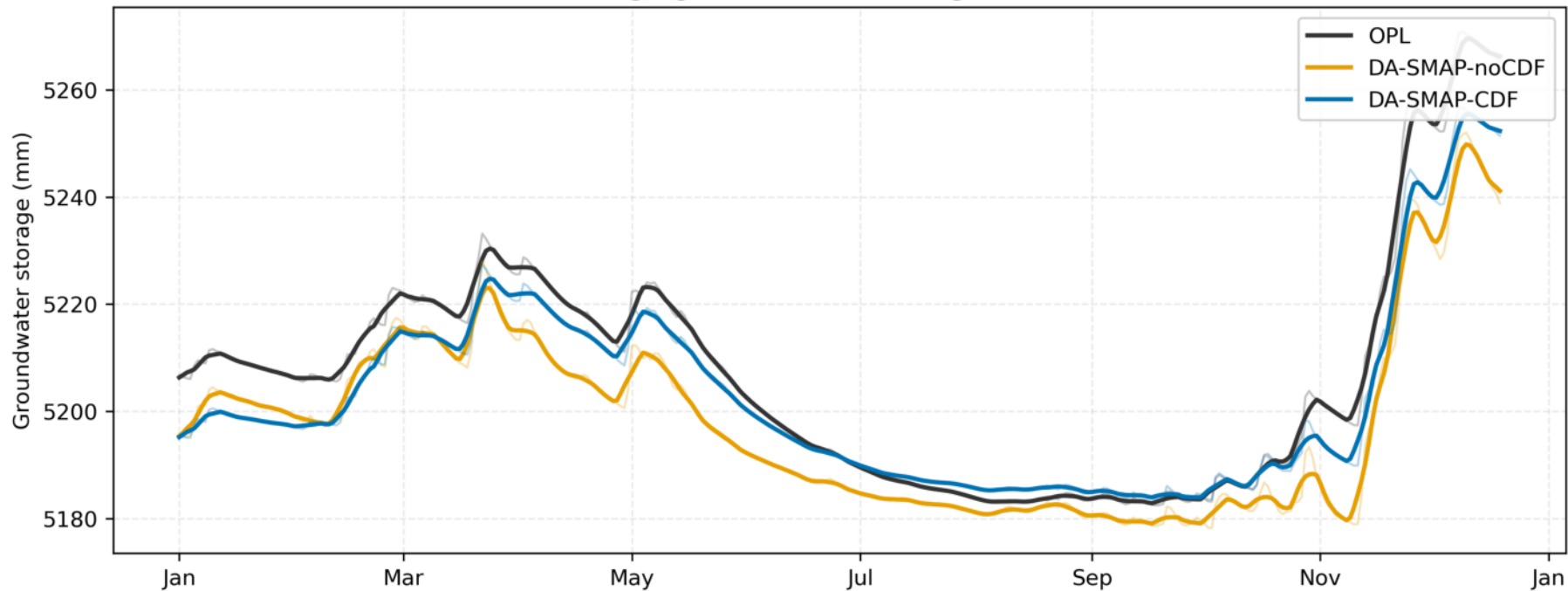


Source file: groundwater_storage_differences_2016.png

Impact of SMAP assimilation on groundwater storage.

Domain-average groundwater storage time series — 2016

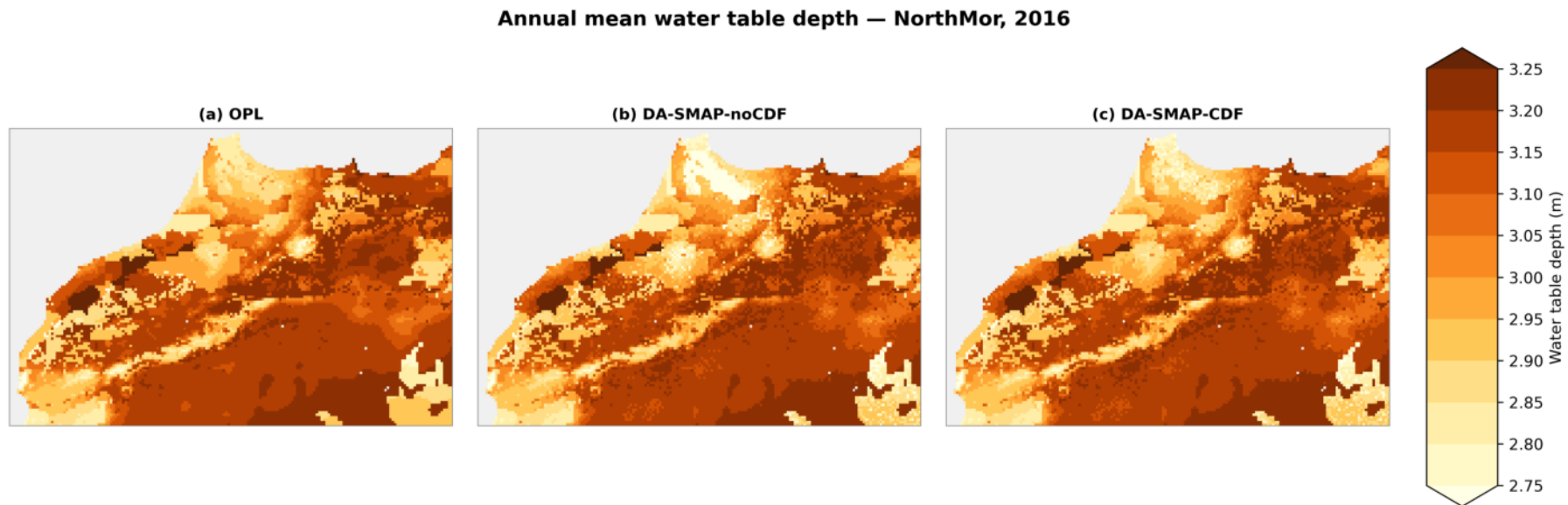
Domain-average groundwater storage — NorthMor, 2016



Source file: groundwater_storage_domain_mean_timeseries_2016.png

Daily domain-averaged timeseries of groundwater storage.

Annual mean water table depth — NorthMor, 2016

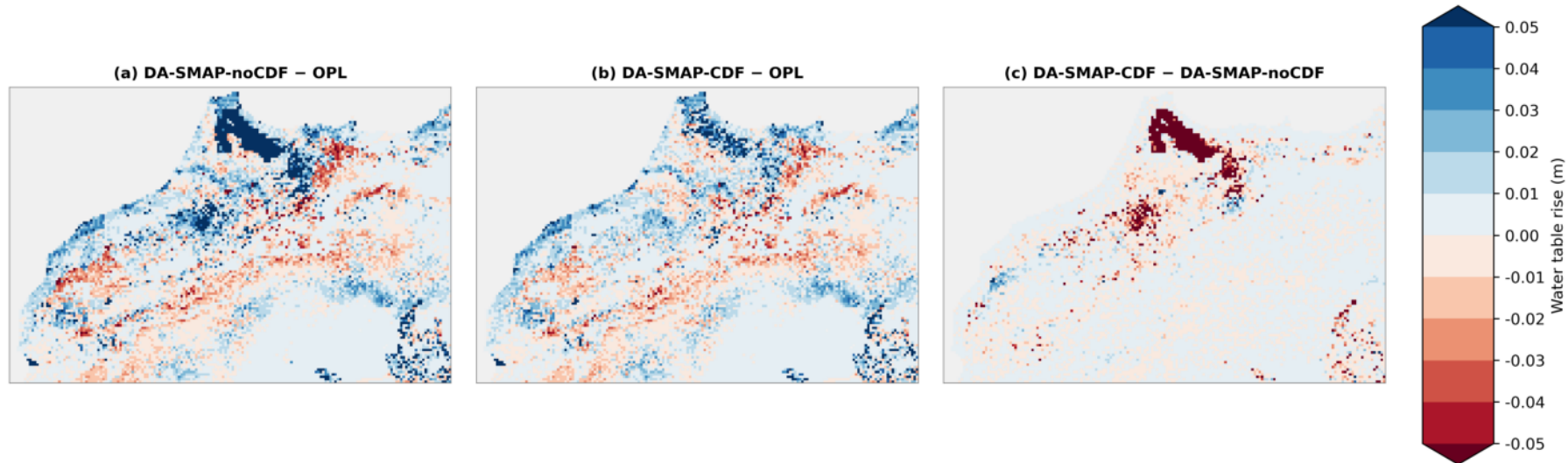


Source file: water_table_depth_annual_mean_2016.png

Annual mean spatial distribution of water table depth.

Water table depth response to assimilation — 2016

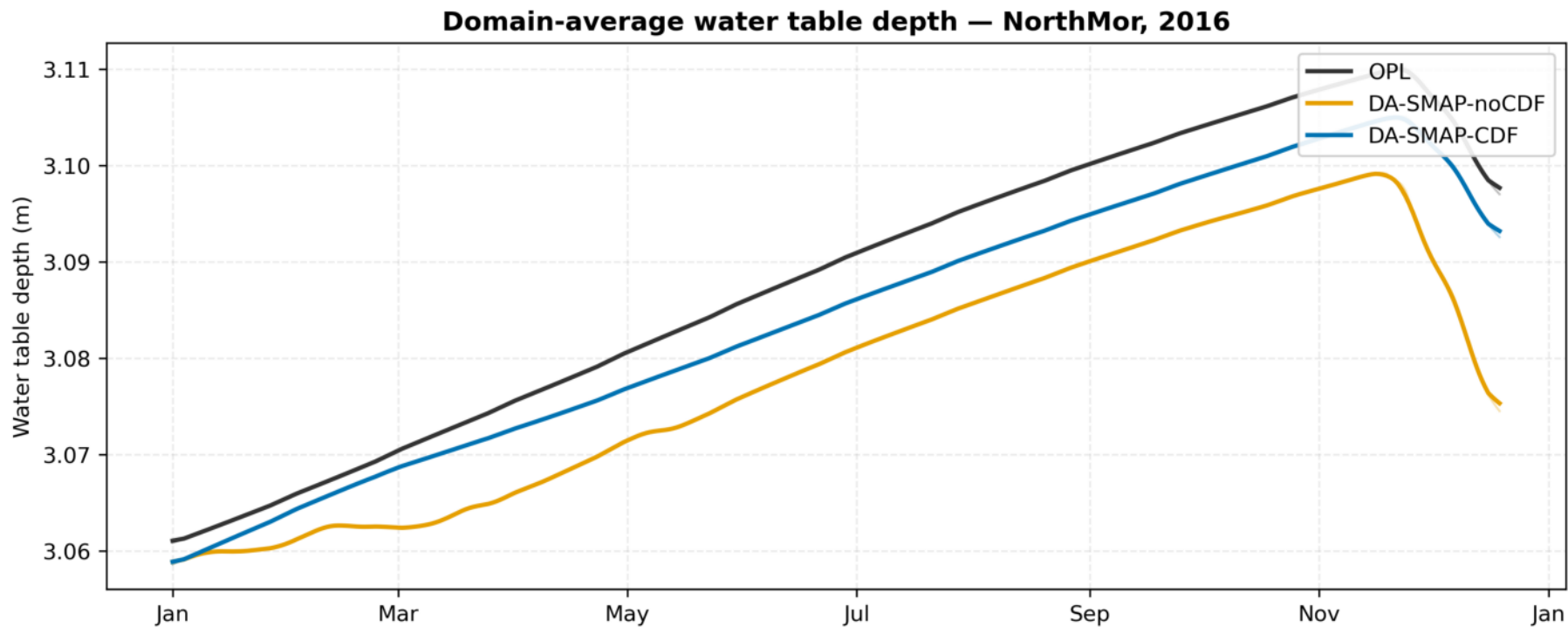
Water table depth response to assimilation — 2016



Source file: water_table_depth_differences_2016.png

Impact of SMAP assimilation on water table depth.

Domain-average water table depth time series — 2016

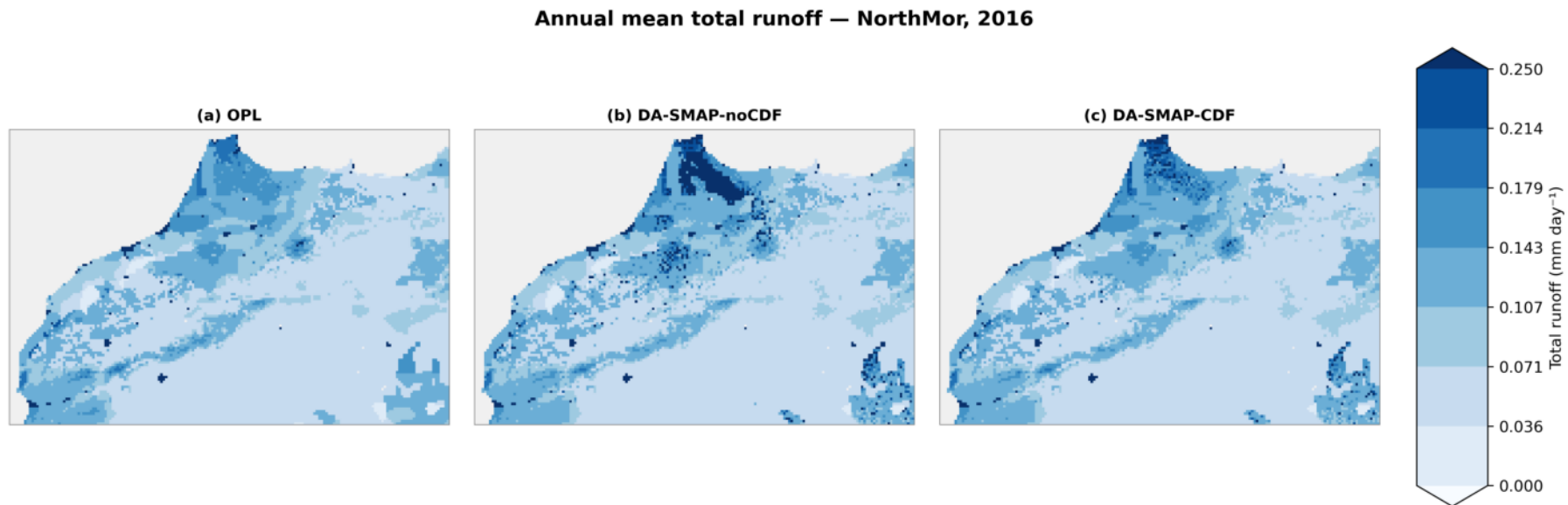


Source file: water_table_depth_domain_mean_timeseries_2016.png

Daily domain-averaged timeseries of water table depth.

9. Runoff and baseflow response

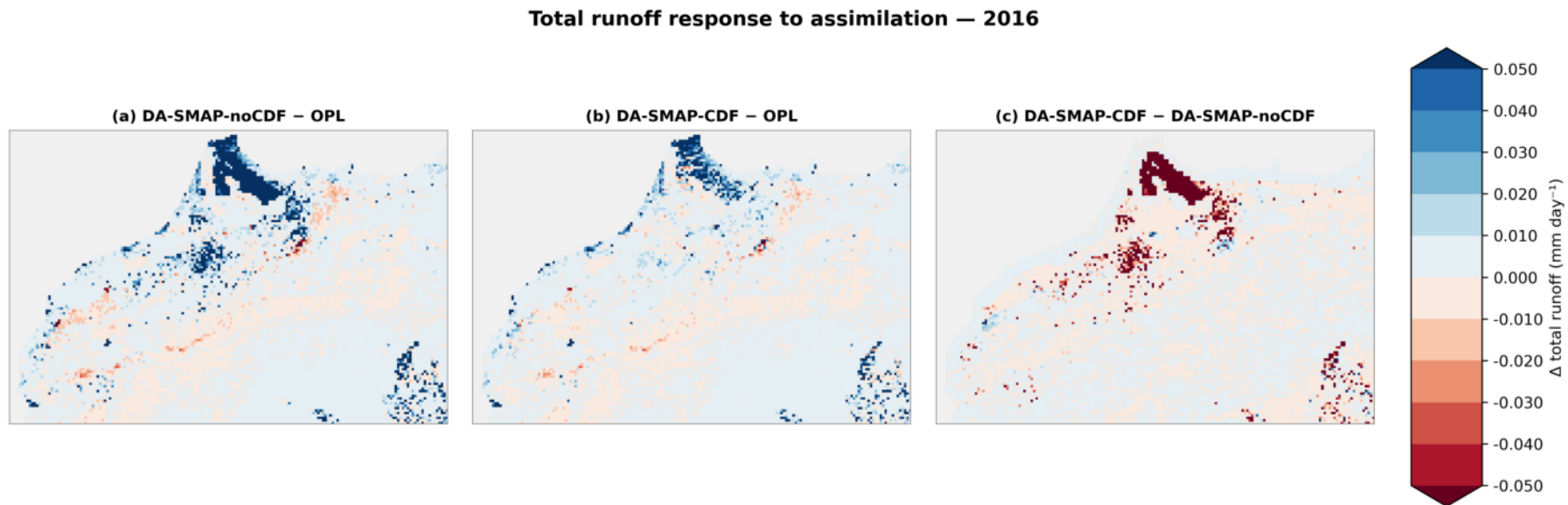
Annual mean total runoff — NorthMor, 2016



Source file: total_runoff_annual_mean_2016.png

Annual mean spatial distribution of total runoff.

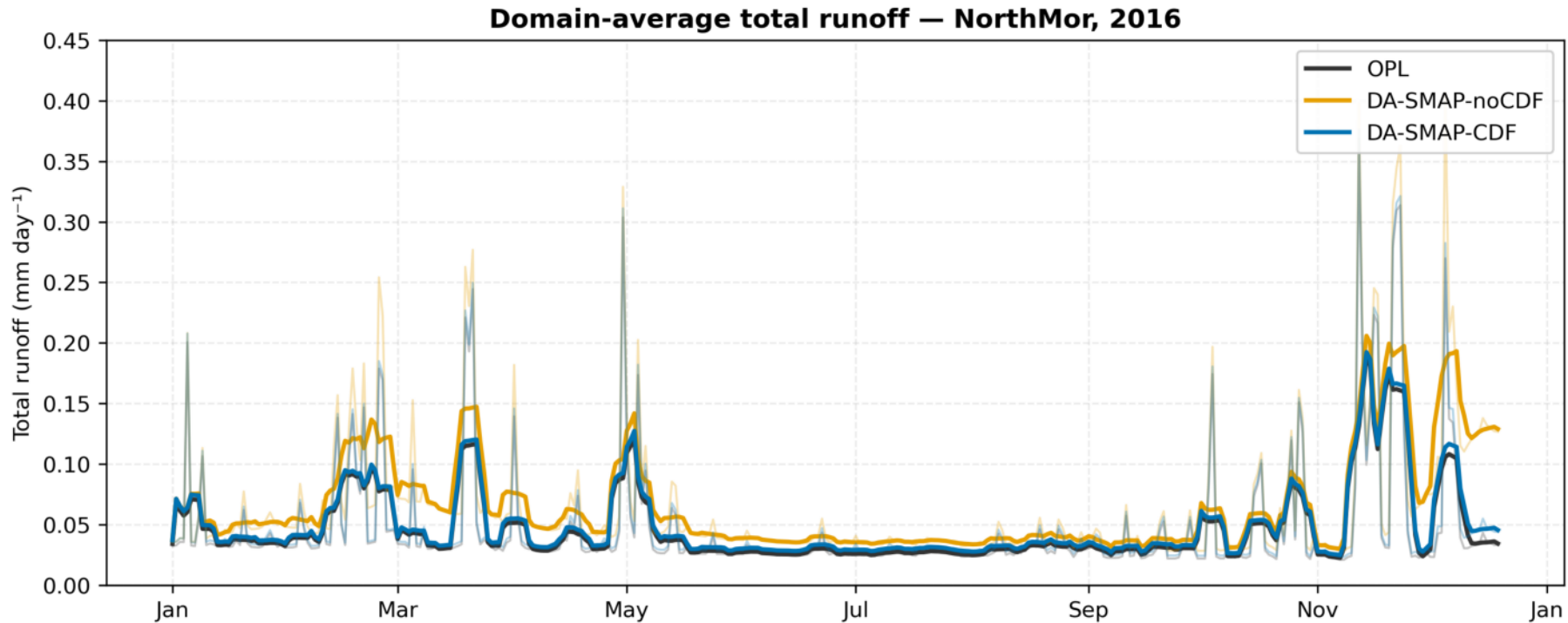
Total runoff response to assimilation — 2016



Source file: total_runoff_differences_2016.png

Impact of SMAP assimilation on total runoff.

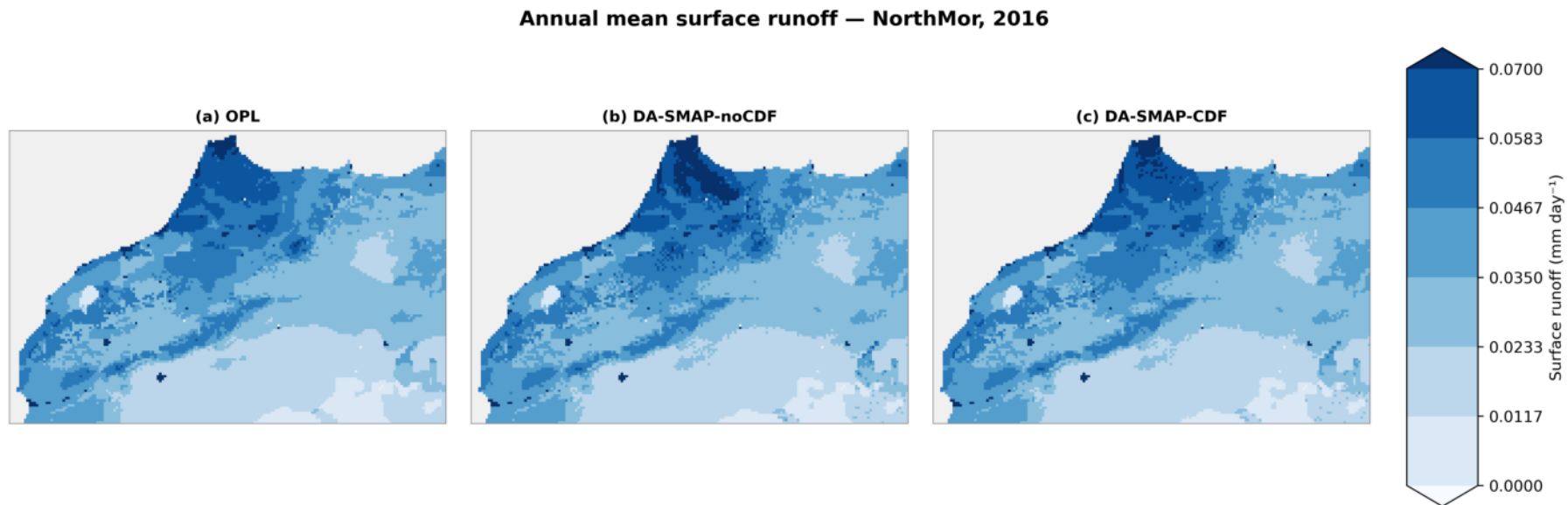
Domain-average total runoff time series — 2016



Source file: total_runoff_domain_mean_timeseries_2016.png

Daily domain-averaged timeseries of total runoff.

Annual mean surface runoff — NorthMor, 2016

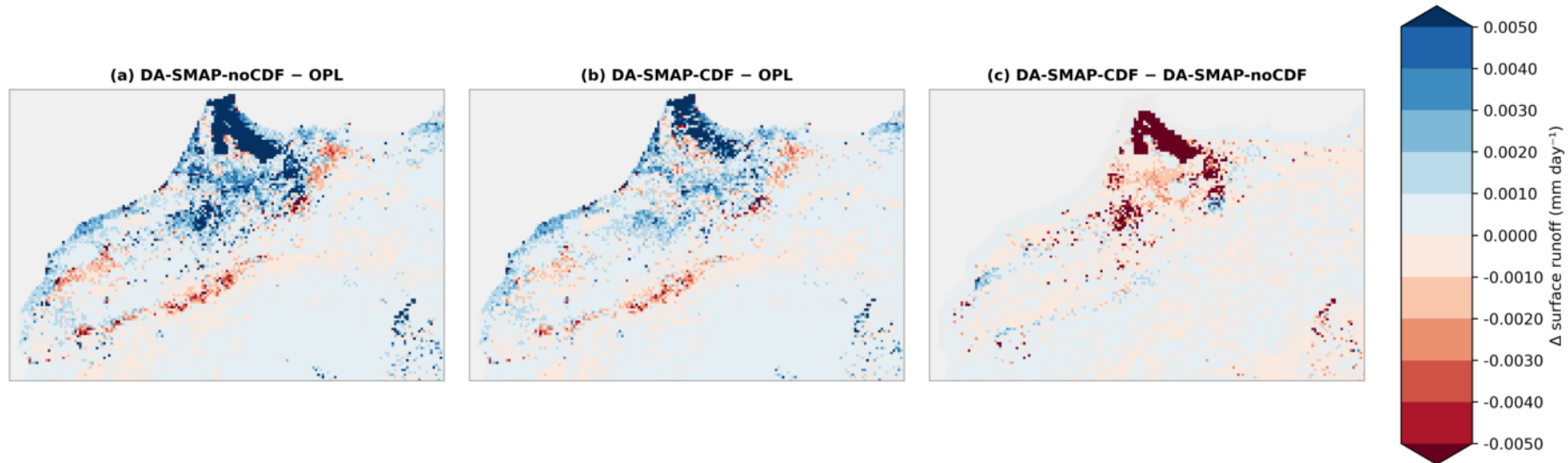


Source file: surface_runoff_annual_mean_2016.png

Annual mean spatial distribution of surface runoff.

Surface runoff response to assimilation — 2016

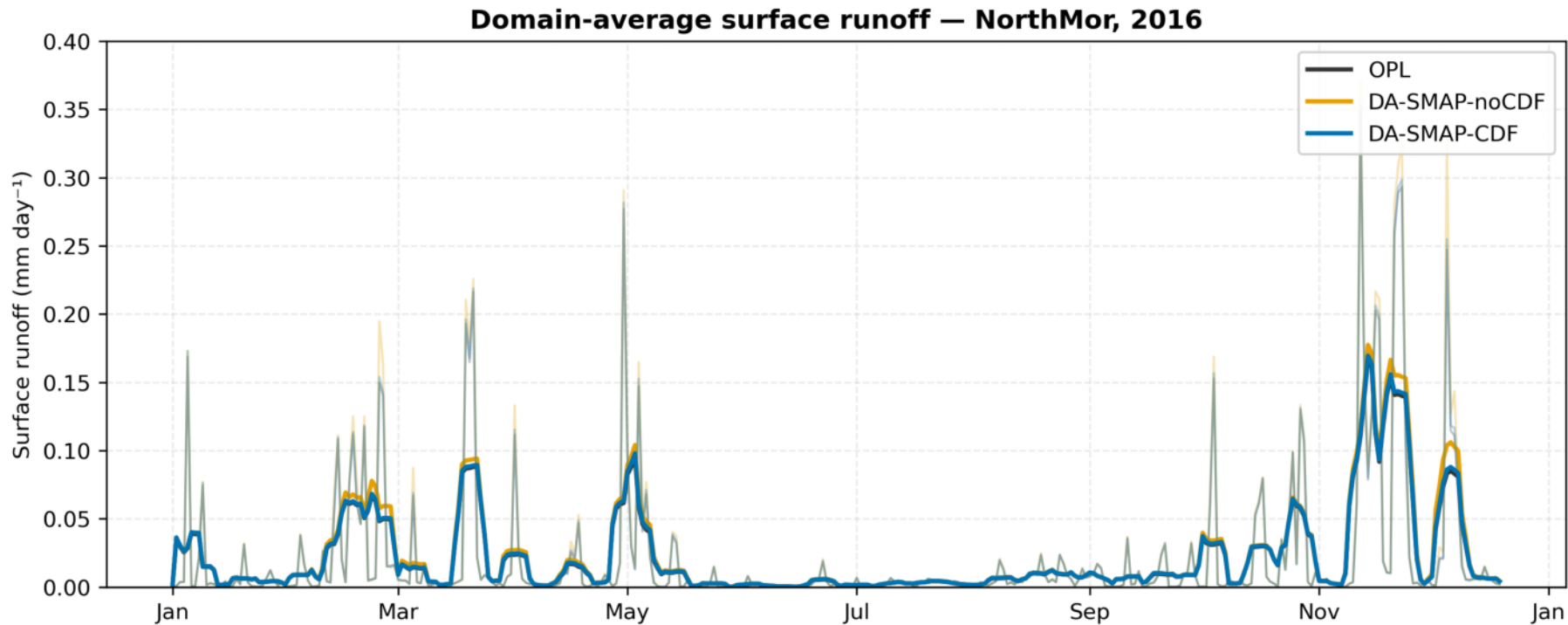
Surface runoff response to assimilation — 2016



Source file: surface_runoff_differences_2016.png

Impact of SMAP assimilation on surface runoff.

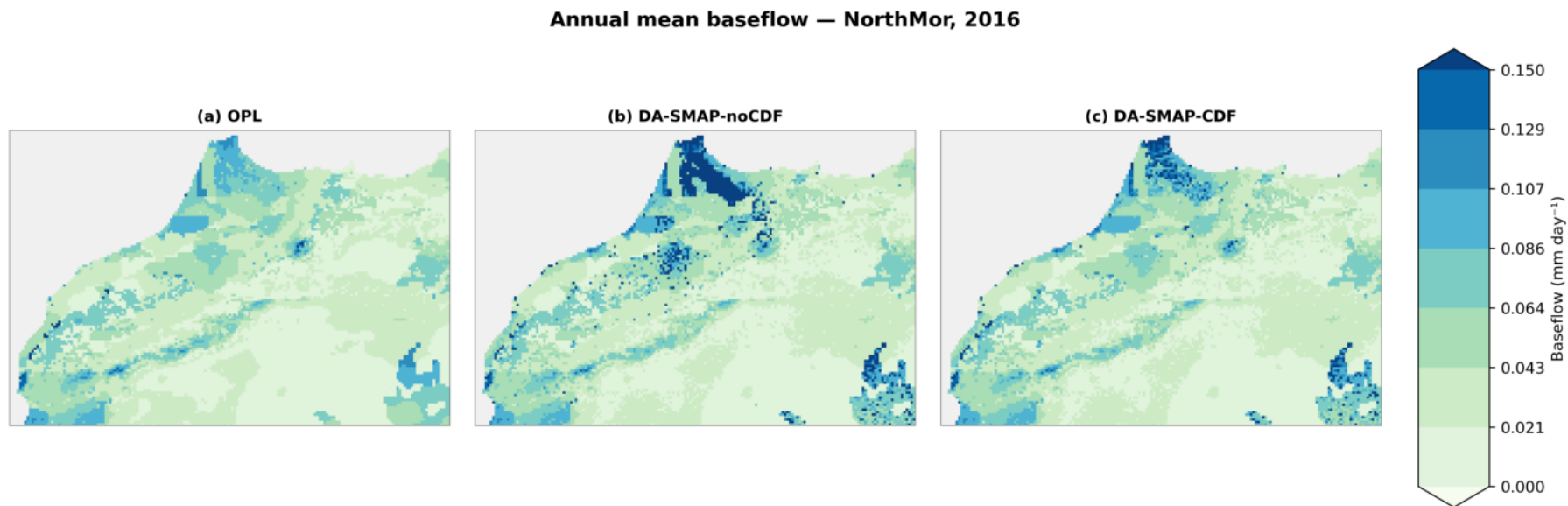
Domain-average surface runoff time series — 2016



Source file: surface_runoff_domain_mean_timeseries_2016.png

Daily domain-averaged timeseries of surface runoff.

Annual mean baseflow — NorthMor, 2016

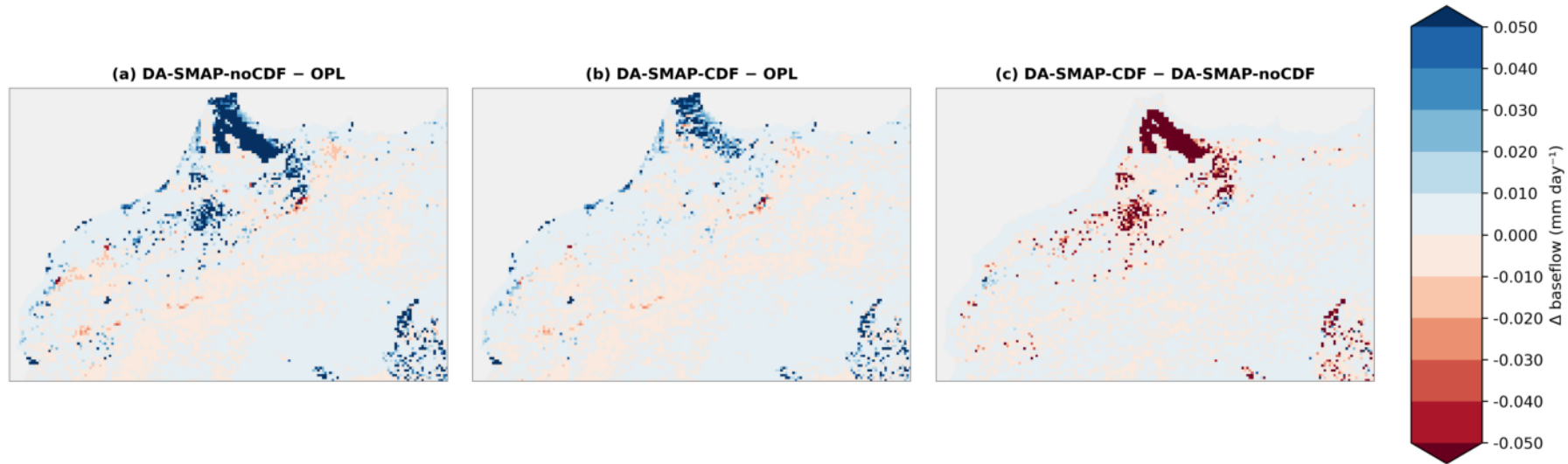


Source file: baseflow_annual_mean_2016.png

Annual mean spatial distribution of baseflow.

Baseflow response to assimilation — 2016

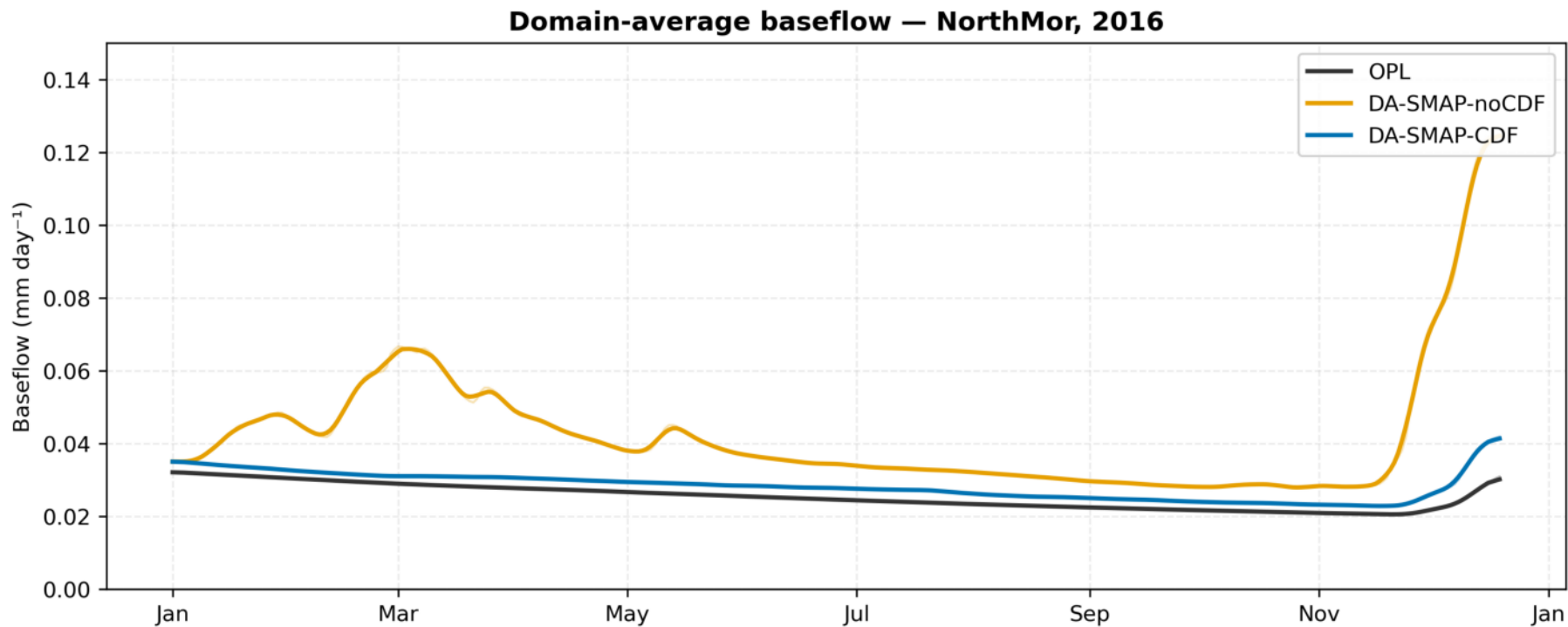
Baseflow response to assimilation — 2016



Source file: baseflow_differences_2016.png

Impact of SMAP assimilation on baseflow.

Domain-average baseflow time series — 2016



Source file: baseflow_domain_mean_timeseries_2016.png

Daily domain-averaged timeseries of baseflow.

10. Streamflow & External validation

Streamflow and External validation figures were not generated.

This typically happens because HyMAP routing outputs are not yet available, or the necessary in-situ observations have not been provided to the framework.

11. Scientific checklist

- [] Are SMAP observations spatially well distributed?
- [] Are there spatial gaps in the assimilation coverage?
- [] Is the monthly observation availability realistic?
- [] Are the noCDF and CDF observation counts consistent?
- [] Are innovations mostly positive, negative, or spatially mixed?
- [] Are increments physically coherent with the sign of innovations?
- [] Is the increment distribution centred close to zero?
- [] Does CDF-matching reduce systematic innovation bias?
- [] Are wet-season and dry-season increments different?
- [] Are there regions with unrealistic or noisy increments?
- [] Are spread diagnostics consistent enough for interpretation?
- [] Which assimilation figures should be kept for the paper?
- [] Which assimilation figures should go to supplementary material?
- [] Which figures need revised colorbar limits, titles, or captions?